

gocachemark - Go Cache Benchmark Report @ 2026-01-02 19:19:28 EST

[gocachemark](#) · darwin/arm64 · 16 CPUs · go1.25.5

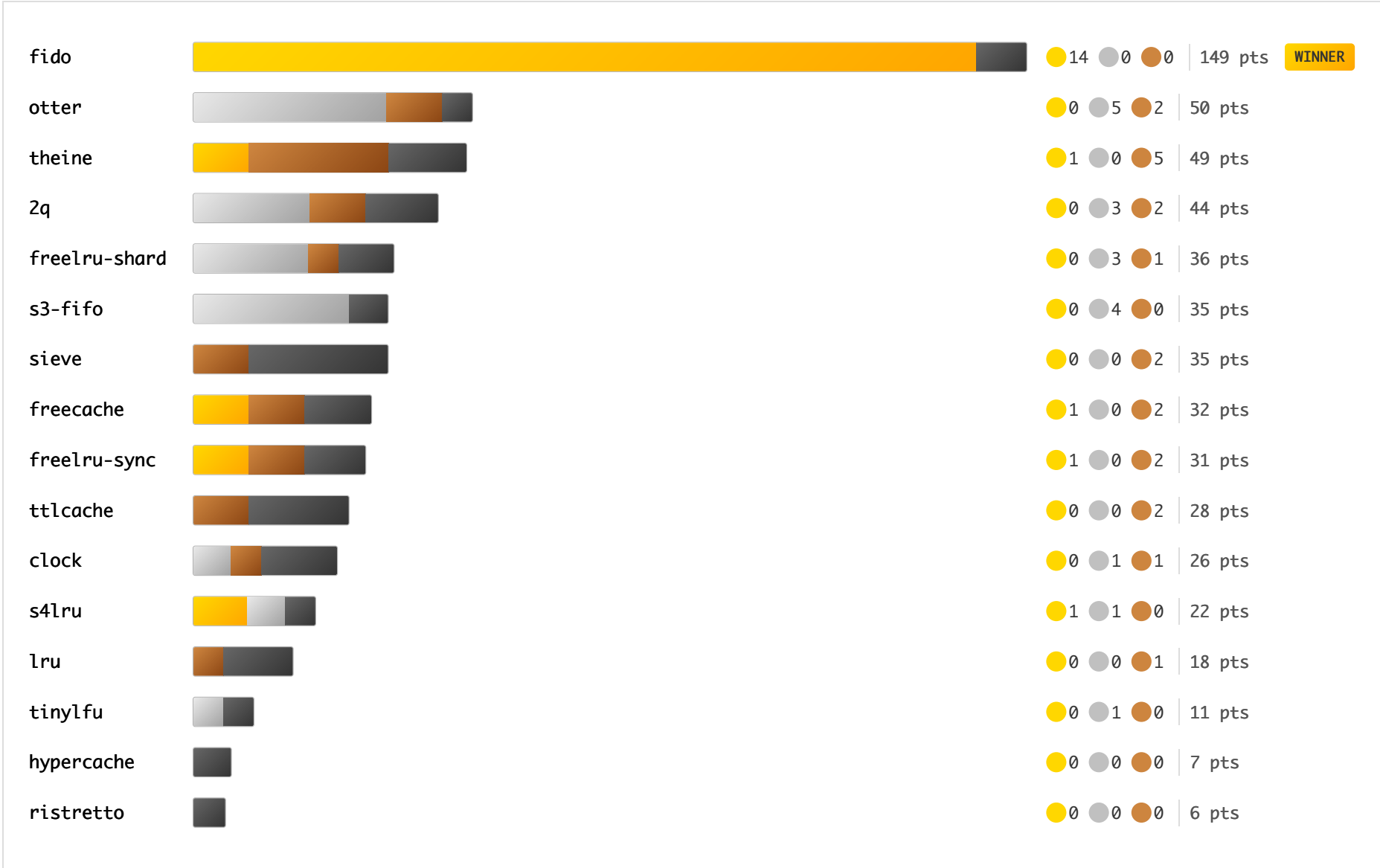
2026-01-02 19:19:28 EST gocachemark

Table of Contents

- [Medal Standings](#)
 - [Medals by Category](#)
 - [Hit Rate Benchmarks](#)
 - [Latency Benchmarks](#)
 - [Throughput Benchmarks](#)
 - [Memory Benchmarks](#)
-

Medal Standings

Ranked by total points earned across all benchmarks



Medals by Category

Hit Rate — Winner: **fido**

fido	● 6
s4lru	● 1 ● 1
theine	● 1 ● 2
freecache	● 1
s3-fifo	● 4
2q	● 3 ● 2
tinylfu	● 1
sieve	● 2
freelru-sync	● 1
ttlcache	● 1
lru	● 1
clock	● 1
otter	● 1

Benchmark	Gold	Silver	Bronze
CDN	fido	s3-fifo	2q
Meta	fido	s3-fifo	theine
Zipf	theine	tinylfu	otter
Twitter	fido	2q	sieve
Wikipedia	fido	s3-fifo	theine
Thesios Block	s4lru	2q	sieve
Thesios File	fido	s4lru	lru, ttlcache, freelru-sync
IBM Docker	freecache	2q	clock
Tencent Photo	fido	s3-fifo	2q

Latency — Winner: **fido**

fido	● 3
otter	● 1 ● 1
clock	● 1
freelru-shard	● 1
freelru-sync	● 1
freecache	● 1

Benchmark	Gold	Silver	Bronze
String Keys	fido	clock	freelru-sync
Int Keys	fido	freelru-shard	otter
GetOrSet	fido	otter	freecache

Throughput — Winner: **fido**

fido	● 5
otter	● 3
freelru-shard	● 2
theine	● 3
freecache	● 1
ttlcache	● 1

Benchmark	Gold	Silver	Bronze
String Get	fido	otter	theine
String Set	fido	freelru-shard	freecache
Int Get	fido	otter	theine
Int Set	fido	freelru-shard	ttlcache
GetOrSet	fido	otter	theine

Memory — Winner: **freelru-sync**

freelru-sync	● 1
otter	● 1
freelru-shard	● 1

Benchmark	Gold	Silver	Bronze
Overhead	freelru-sync	otter	freelru-shard

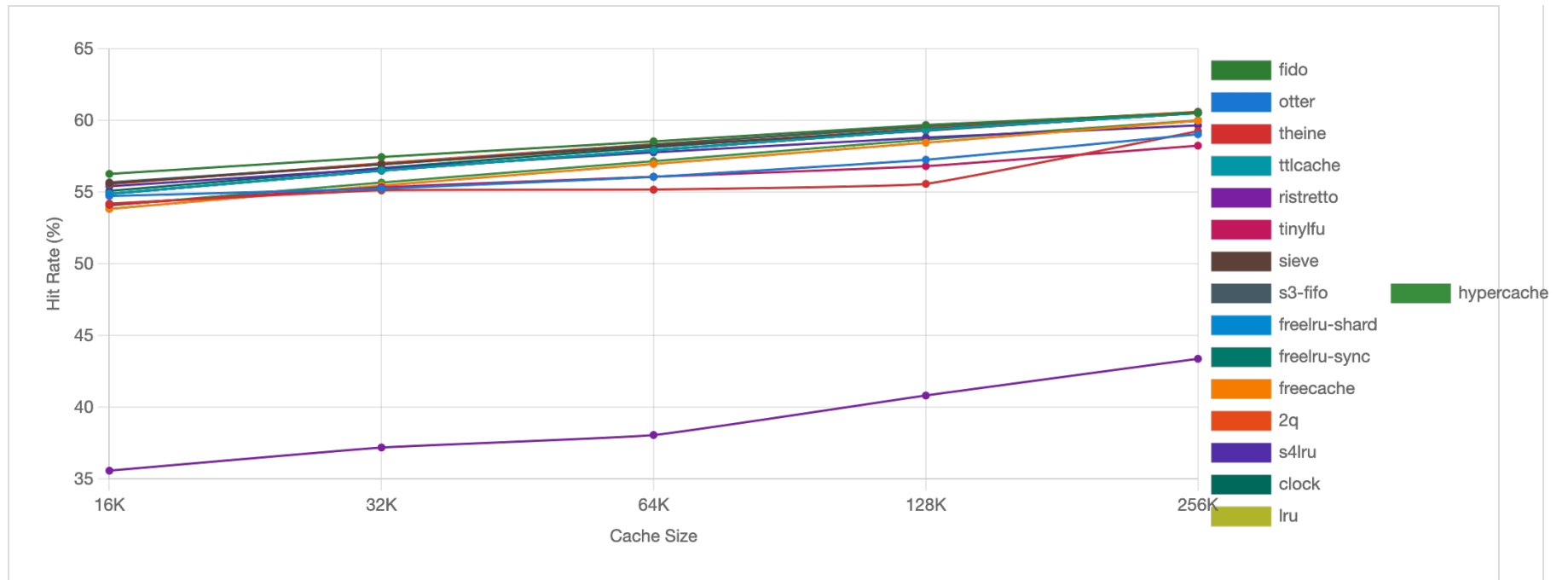
Hit Rate Benchmarks



Higher hit rates mean better cache effectiveness. Tests measure how often requested items are found in the cache across different cache sizes.

CDN Production Trace

2M operations, ~768K unique keys. Source: [libCacheSim](#)



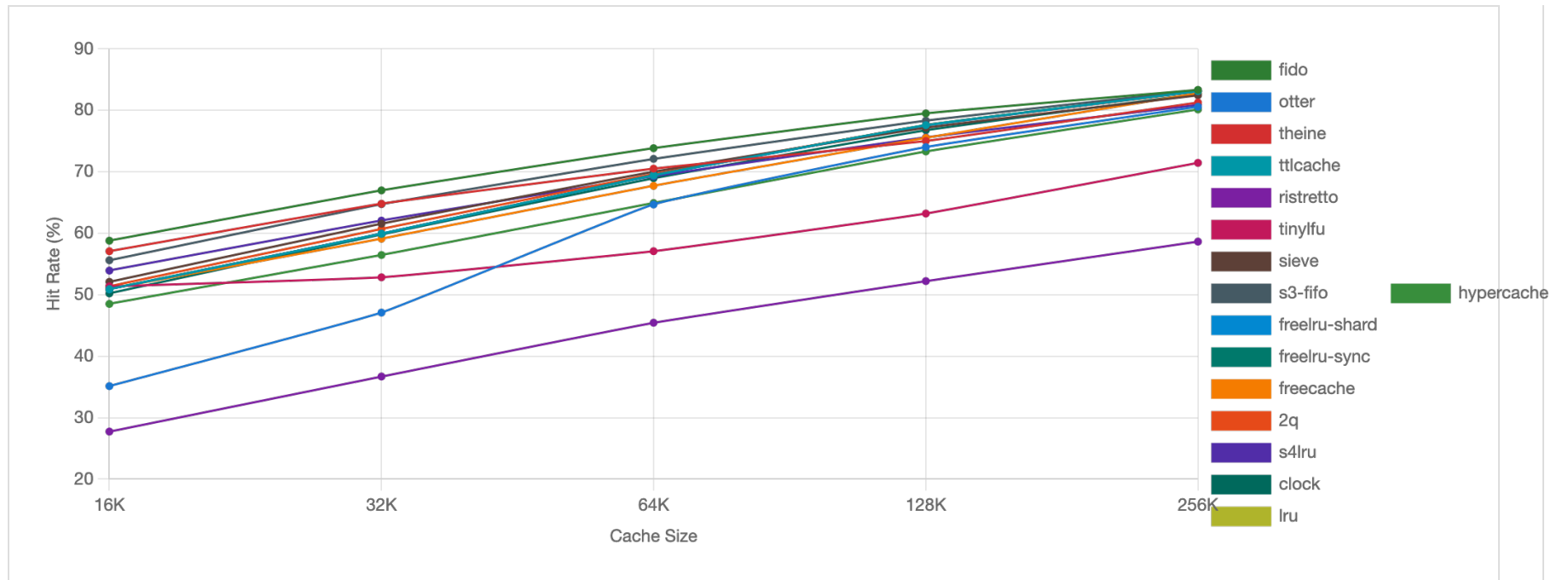
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	56.282%	57.457%	58.553%	59.694%	60.554%	58.508%
s3-fifo	55.690%	56.948%	58.327%	59.611%	60.554%	58.226%
2q	55.575%	57.019%	58.349%	59.559%	60.613%	58.223%
sieve	55.634%	56.947%	58.248%	59.298%	60.587%	58.143%
clock	55.093%	56.675%	58.138%	59.422%	60.566%	57.979%
freelru-sync	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%
lru	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%
ttlcache	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	54.886%	56.507%	57.940%	59.344%	60.532%	57.842%
s4lru	55.423%	56.644%	57.789%	58.820%	59.674%	57.670%
hypercache	54.083%	55.685%	57.168%	58.675%	60.011%	57.124%
freecache	53.849%	55.478%	56.976%	58.468%	59.998%	56.954%
otter	54.743%	55.257%	56.081%	57.268%	59.053%	56.480%
tinylfu	54.139%	55.350%	56.081%	56.825%	58.248%	56.128%
theine	54.211%	55.141%	55.187%	55.590%	59.259%	55.878%
ristretto	35.588%	37.199%	38.065%	40.831%	43.394%	39.015%

Meta KVCache Production Trace

5M operations. Source: [CacheLib](#)



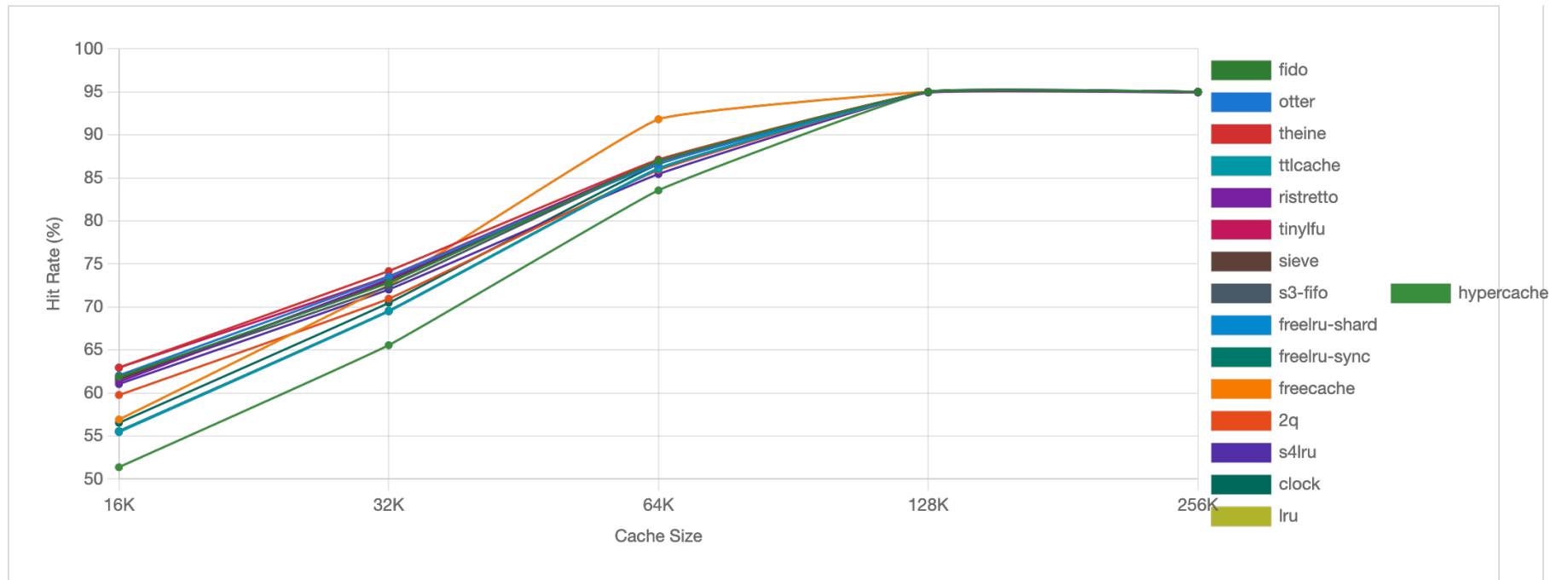
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	58.798%	66.991%	73.854%	79.505%	83.330%	72.495%
s3-fifo	55.609%	64.742%	72.101%	78.334%	83.275%	70.812%
theine	57.082%	64.809%	70.508%	75.009%	81.257%	69.733%
sieve	52.098%	61.564%	70.016%	77.166%	82.451%	68.659%
2q	51.375%	60.714%	69.647%	77.533%	83.073%	68.469%
s4lru	53.943%	62.068%	69.274%	75.591%	80.867%	68.349%
freelru-sync	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%
lru	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%

Cache	16K	32K	64K	128K	256K	Avg
ttlcache	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%
freelru-shard	50.895%	59.941%	69.400%	77.509%	83.114%	68.172%
clock	50.228%	59.802%	68.951%	76.758%	82.677%	67.683%
freecache	50.947%	59.120%	67.726%	75.525%	82.764%	67.216%
hypercache	48.525%	56.469%	64.921%	73.303%	80.145%	64.672%
otter	35.126%	47.096%	64.687%	74.017%	80.582%	60.302%
tinylfu	51.354%	52.830%	57.071%	63.202%	71.448%	59.181%
ristretto	27.730%	36.689%	45.437%	52.213%	58.647%	44.143%

Zipf Synthetic Trace

alpha=0.8, 2M operations, 100K keyspace



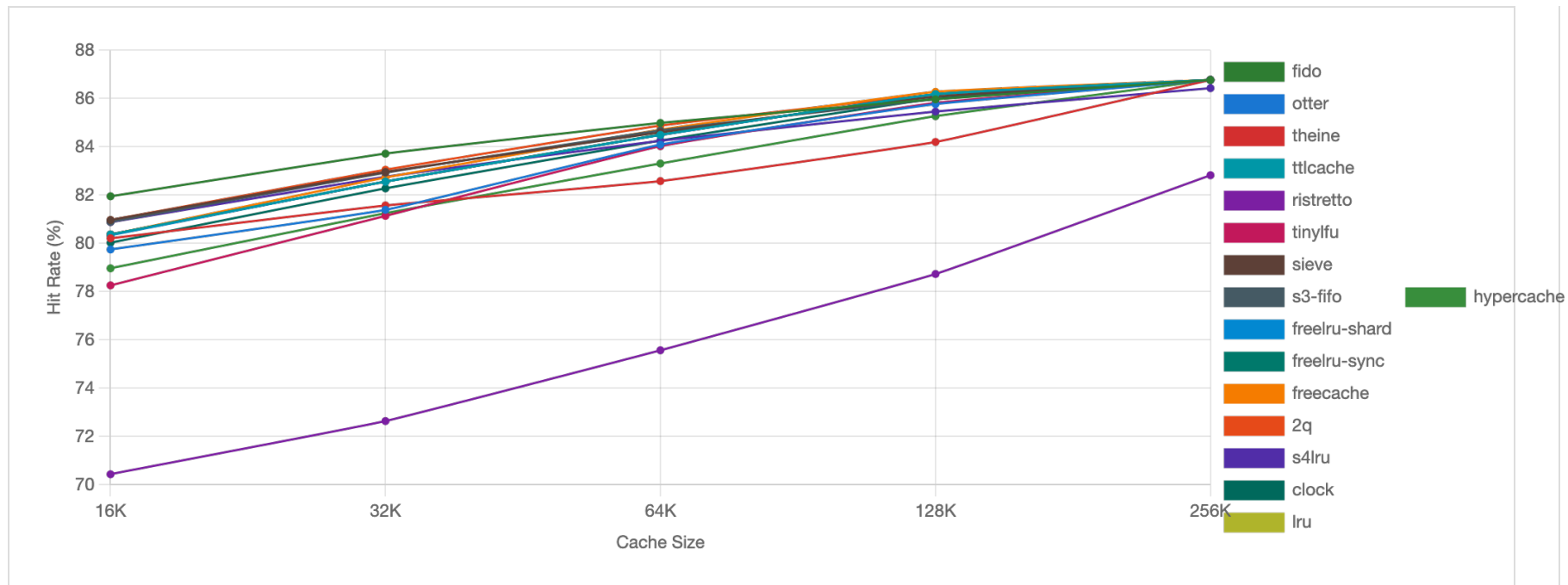
Results Table

Cache	16K	32K	64K	128K	256K	Avg
theine	62.960%	74.175%	87.126%	95.014%	95.014%	82.858%
tinylfu	62.964%	73.573%	86.917%	95.014%	95.014%	82.696%
otter	62.053%	73.496%	86.629%	95.014%	95.014%	82.441%
fido	61.896%	72.833%	86.965%	95.014%	95.014%	82.344%
sieve	61.558%	73.071%	86.998%	95.014%	95.014%	82.331%
ristretto	61.314%	73.246%	87.006%	94.912%	94.935%	82.283%
freecache	56.915%	72.559%	91.823%	95.014%	95.014%	82.265%
s3-fifo	61.740%	72.401%	86.912%	95.014%	95.014%	82.216%

Cache	16K	32K	64K	128K	256K	Avg
s4lru	61.043%	72.030%	85.442%	95.014%	95.014%	81.709%
2q	59.772%	70.969%	85.935%	95.014%	95.014%	81.341%
clock	56.541%	70.476%	86.600%	95.014%	95.014%	80.729%
freelru-sync	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
ttlcache	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
lru	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
freelru-shard	55.472%	69.499%	86.069%	95.014%	95.014%	80.214%
hypercache	51.380%	65.560%	83.560%	95.014%	95.014%	78.106%

Twitter Production Cache Trace

2M operations, cluster001+cluster052. Source: [Twitter Cache Trace](#)



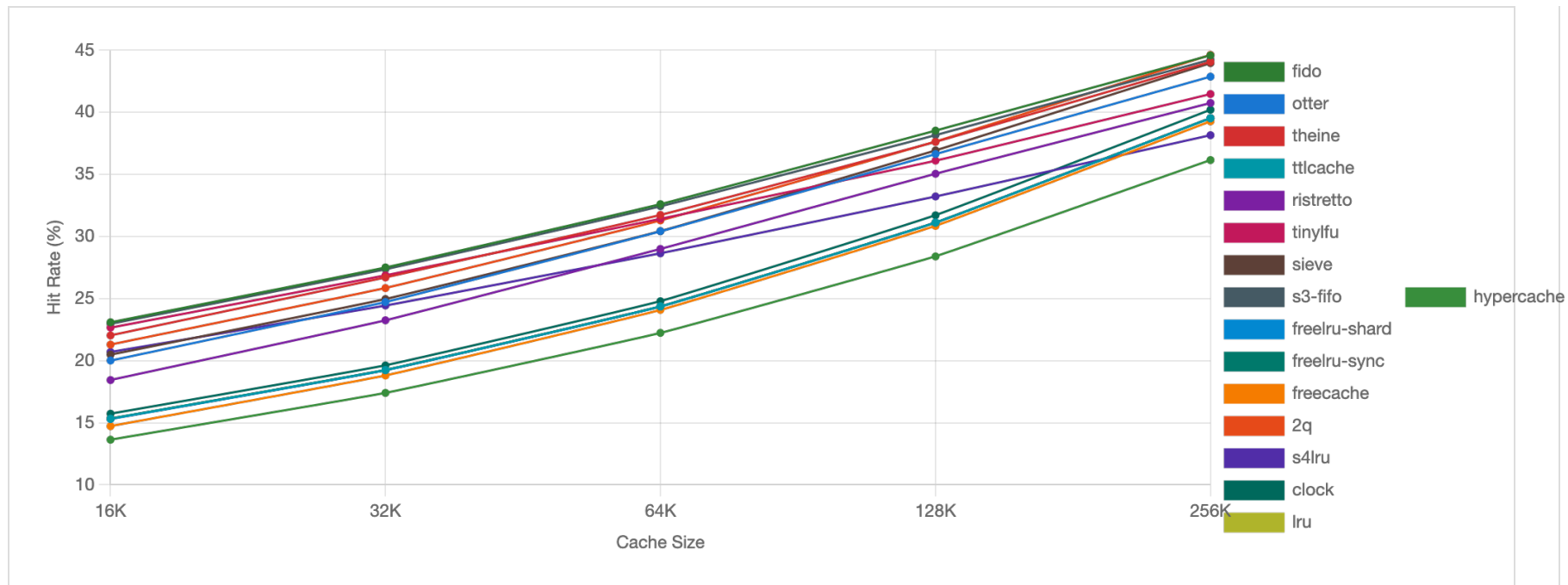
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	81.946%	83.717%	84.991%	85.969%	86.773%	84.679%
2q	80.970%	83.050%	84.875%	86.166%	86.773%	84.367%
sieve	80.971%	82.965%	84.599%	86.077%	86.773%	84.277%
s3-fifo	80.891%	82.933%	84.684%	85.969%	86.773%	84.250%
freecache	80.357%	82.733%	84.708%	86.278%	86.773%	84.170%
freelru-sync	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%
lru	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%
ttlcache	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	80.341%	82.561%	84.492%	86.179%	86.772%	84.069%
s4lru	80.883%	82.760%	84.250%	85.464%	86.425%	83.957%
clock	80.030%	82.285%	84.257%	85.997%	86.773%	83.868%
otter	79.745%	81.390%	84.087%	85.769%	86.773%	83.553%
tinylfu	78.260%	81.144%	84.030%	85.820%	86.772%	83.205%
hypercache	78.974%	81.245%	83.314%	85.275%	86.773%	83.116%
theine	80.207%	81.569%	82.582%	84.198%	86.773%	83.066%
ristretto	70.443%	72.639%	75.567%	78.730%	82.822%	76.040%

Wikipedia CDN Upload Trace

2M operations, upload.wikimedia.org. Source: [libCacheSim](#)



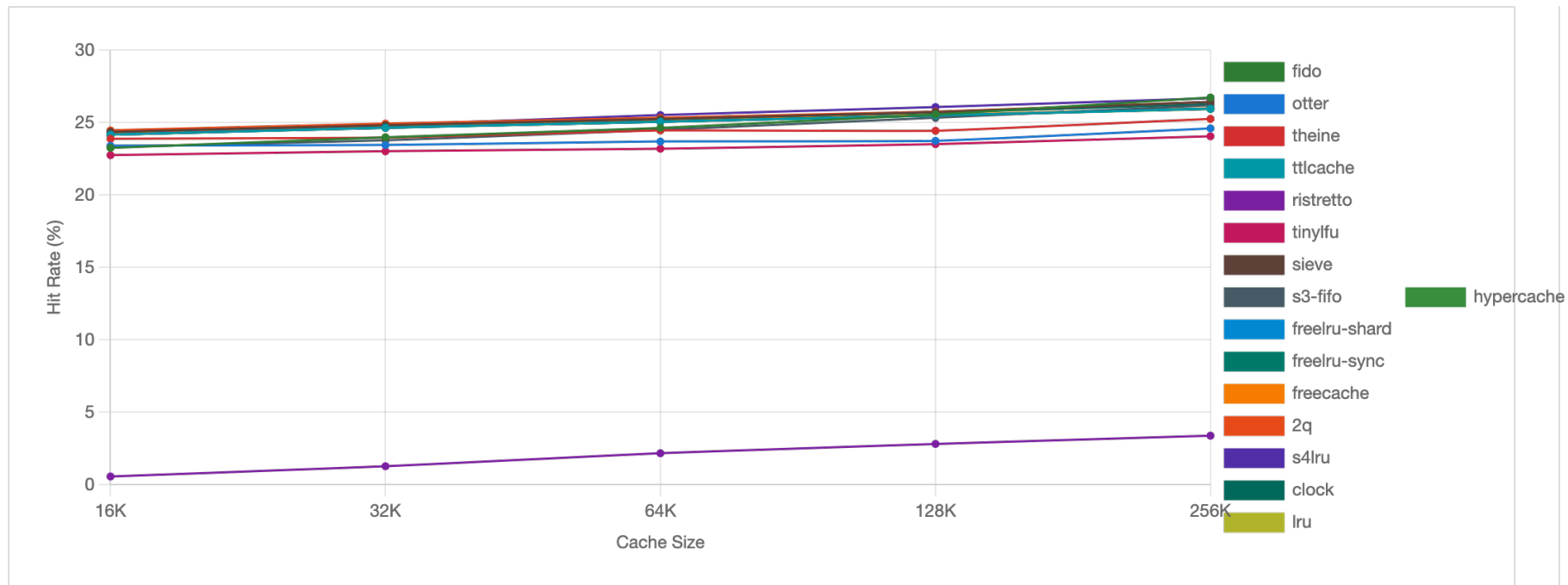
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	23.093%	27.506%	32.610%	38.534%	44.601%	33.269%
s3-fifo	22.970%	27.374%	32.441%	38.173%	44.250%	33.042%
theine	22.035%	26.705%	31.733%	37.621%	44.056%	32.430%
2q	21.286%	25.850%	31.285%	37.635%	44.619%	32.135%
tinylfu	22.663%	26.880%	31.434%	36.112%	41.483%	31.714%
sieve	20.476%	24.956%	30.430%	36.932%	43.963%	31.351%
otter	20.011%	24.734%	30.423%	36.636%	42.884%	30.938%
ristretto	18.442%	23.249%	29.002%	35.054%	40.750%	29.299%

Cache	16K	32K	64K	128K	256K	Avg
s4lru	20.697%	24.438%	28.654%	33.215%	38.155%	29.032%
clock	15.725%	19.617%	24.792%	31.711%	40.196%	26.408%
freelru-sync	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
ttlcache	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
lru	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
freelru-shard	15.299%	19.233%	24.364%	31.117%	39.529%	25.908%
freecache	14.732%	18.806%	24.090%	30.856%	39.283%	25.554%
hypercache	13.633%	17.398%	22.230%	28.402%	36.146%	23.562%

Google Thesios I/O Block Trace

400K read operations, ~322K unique blocks. Source: [Tectonic-Shift \(SOSP '24\)](#)



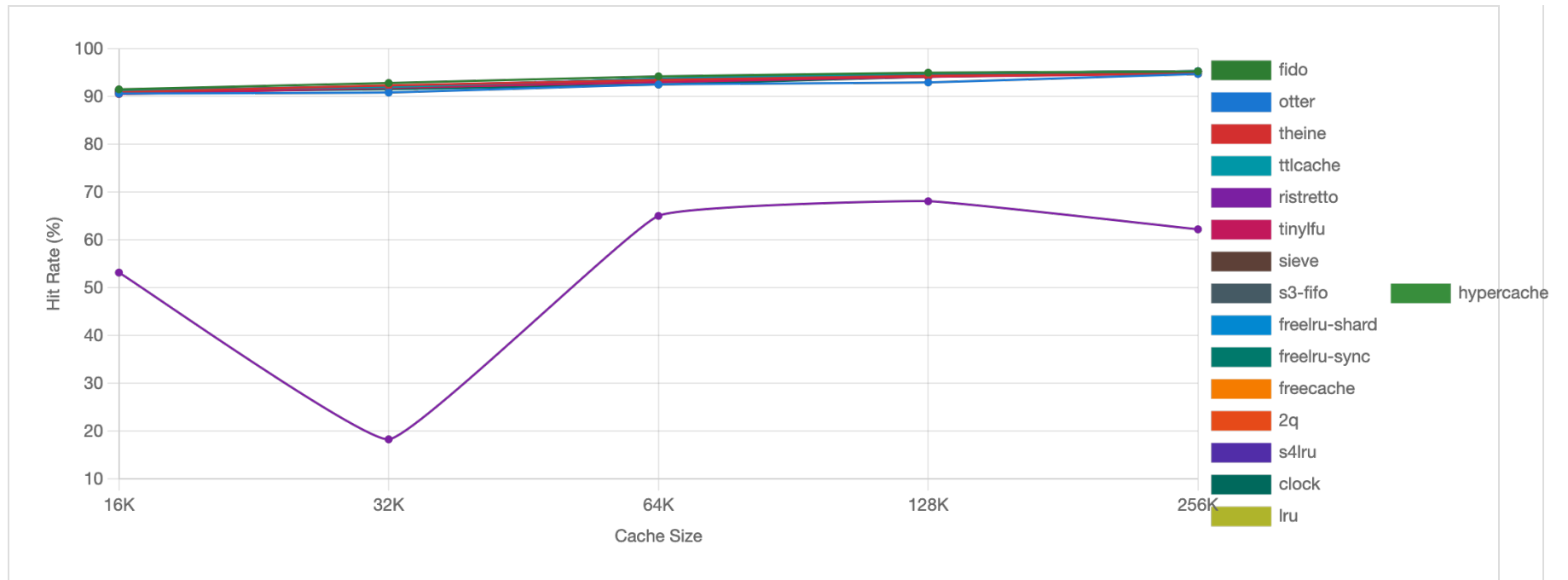
Results Table

Cache	16K	32K	64K	128K	256K	Avg
s4lru	24.191%	24.839%	25.526%	26.082%	26.693%	25.466%
2q	24.467%	24.925%	25.340%	25.760%	26.430%	25.384%
sieve	24.362%	24.818%	25.246%	25.726%	26.428%	25.316%
clock	24.278%	24.756%	25.200%	25.694%	26.280%	25.242%
freecache	24.211%	24.685%	25.103%	25.537%	25.991%	25.105%
freelru-sync	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%
lru	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%
ttlcache	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	24.177%	24.653%	25.067%	25.501%	25.965%	25.072%
hypercache	24.168%	24.640%	25.059%	25.492%	25.940%	25.060%
fido	23.262%	23.995%	24.636%	25.573%	26.738%	24.841%
s3-fifo	23.308%	23.784%	24.505%	25.336%	26.186%	24.624%
theine	23.889%	23.972%	24.462%	24.444%	25.262%	24.406%
otter	23.405%	23.458%	23.699%	23.740%	24.598%	23.780%
tinylfu	22.756%	23.028%	23.198%	23.519%	24.064%	23.313%
ristretto	0.574%	1.279%	2.176%	2.823%	3.392%	2.049%

Google Thesios I/O File Trace

400K read operations, ~46K unique files. Source: [Tectonic-Shift \(SOSP '24\)](#)



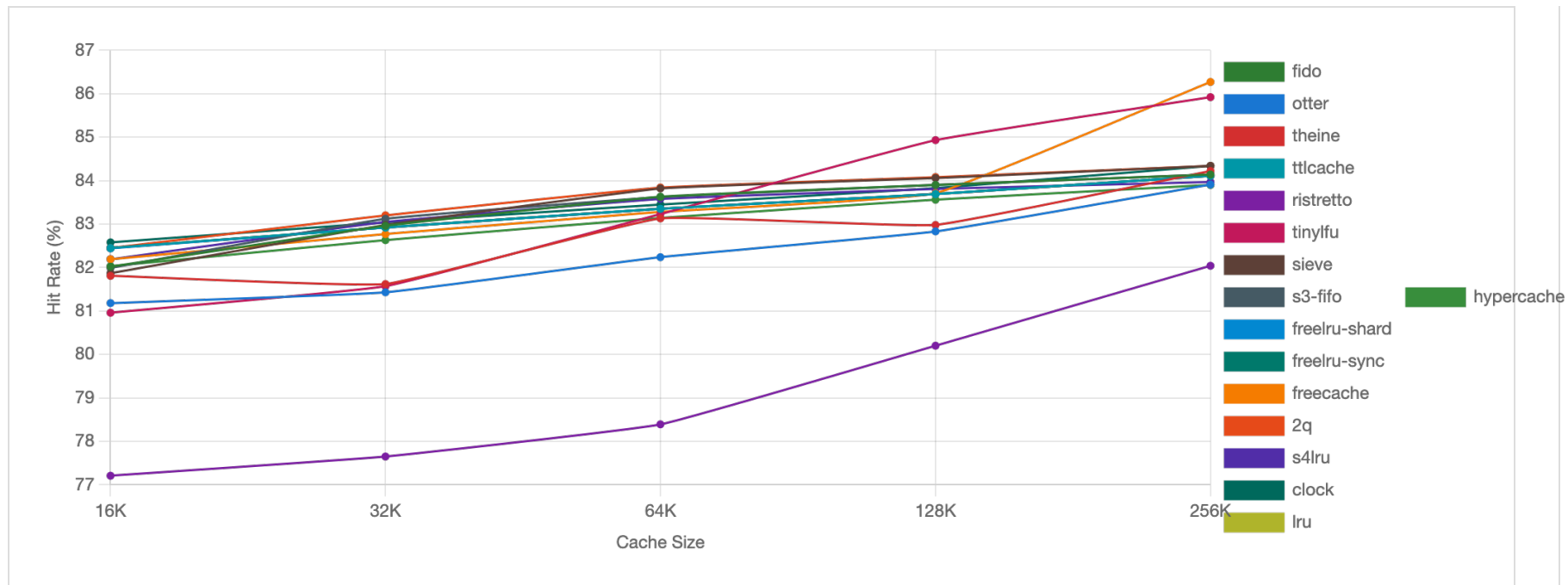
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	91.553%	92.881%	94.249%	95.030%	95.302%	93.803%
s4lru	91.260%	92.211%	93.663%	94.840%	95.310%	93.457%
lru	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
ttlcache	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
freelru-sync	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
freelru-shard	91.291%	92.026%	93.730%	94.849%	95.310%	93.441%
sieve	91.287%	92.041%	93.699%	94.840%	95.260%	93.425%
2q	91.279%	92.094%	93.689%	94.714%	95.290%	93.413%

Cache	16K	32K	64K	128K	256K	Avg
s3-fifo	91.193%	92.181%	93.520%	94.671%	95.211%	93.355%
theine	90.933%	92.347%	93.491%	94.340%	94.784%	93.179%
freecache	91.228%	91.898%	93.194%	94.357%	95.175%	93.170%
hypercache	91.190%	91.827%	93.043%	94.214%	94.912%	93.037%
clock	90.965%	91.559%	92.565%	94.175%	95.170%	92.887%
tinylfu	90.536%	91.832%	93.032%	94.147%	94.779%	92.865%
otter	90.633%	90.872%	92.551%	92.975%	94.743%	92.355%
ristretto	53.203%	18.308%	65.074%	68.132%	62.253%	53.394%

IBM Docker Registry Trace

725K GET requests, ~121K unique URIs. Source: [FAST '18](#)



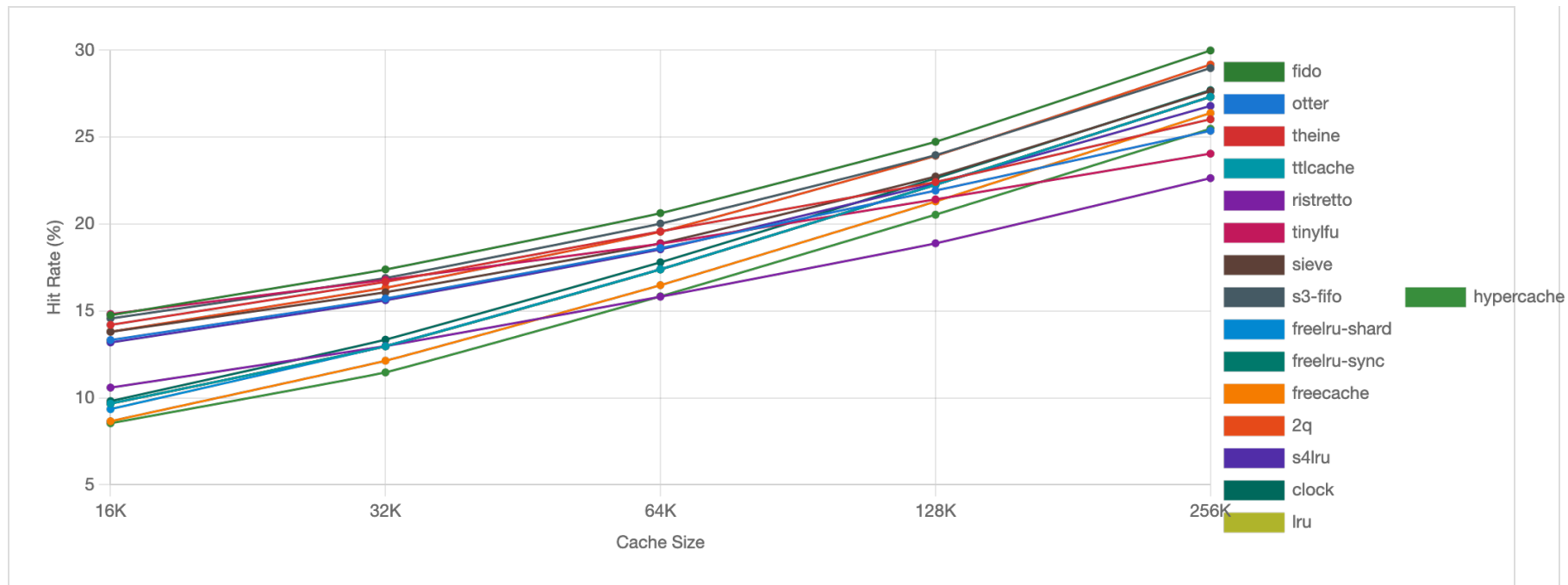
Results Table

Cache	16K	32K	64K	128K	256K	Avg
freecache	82.192%	82.773%	83.280%	83.705%	86.271%	83.644%
2q	82.451%	83.197%	83.840%	84.082%	84.344%	83.583%
clock	82.576%	83.041%	83.446%	83.826%	84.344%	83.446%
sieve	81.868%	82.980%	83.822%	84.064%	84.344%	83.416%
s3-fifo	81.994%	83.124%	83.616%	83.897%	84.141%	83.354%
fido	82.017%	82.962%	83.627%	83.904%	84.141%	83.330%
s4lru	82.189%	83.050%	83.579%	83.810%	83.975%	83.321%
tinylfu	80.956%	81.574%	83.225%	84.927%	85.916%	83.320%

Cache	16K	32K	64K	128K	256K	Avg
lru	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
ttlcache	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
freelru-sync	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
freelru-shard	82.439%	82.920%	83.348%	83.690%	84.110%	83.301%
hypercache	82.026%	82.626%	83.140%	83.559%	83.896%	83.050%
theine	81.812%	81.616%	83.128%	82.980%	84.219%	82.751%
otter	81.177%	81.430%	82.242%	82.828%	83.913%	82.318%
ristretto	77.206%	77.653%	78.394%	80.198%	82.040%	79.098%

Tencent Photo Trace

2M requests, ~1.34M unique photos. Source: [ATC '18](#)



Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	14.755%	17.389%	20.625%	24.738%	29.988%	21.499%
s3-fifo	14.570%	16.903%	20.031%	23.965%	28.982%	20.890%
2q	13.813%	16.336%	19.564%	23.928%	29.182%	20.565%
sieve	13.795%	16.084%	18.895%	22.736%	27.648%	19.831%
theine	14.197%	16.678%	19.578%	22.420%	26.025%	19.780%
s4lru	13.187%	15.624%	18.551%	22.347%	26.802%	19.302%
tinylfu	14.817%	16.786%	18.872%	21.420%	24.051%	19.189%
otter	13.330%	15.720%	18.619%	21.933%	25.370%	18.994%

Cache	16K	32K	64K	128K	256K	Avg
clock	9.811%	13.352%	17.799%	22.631%	27.696%	18.258%
ttlcache	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
lru	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
freelru-sync	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
freelru-shard	9.351%	12.978%	17.399%	22.254%	27.318%	17.860%
freecache	8.663%	12.143%	16.491%	21.298%	26.395%	16.998%
hypercache	8.539%	11.470%	15.835%	20.541%	25.476%	16.372%
ristretto	10.590%	12.984%	15.818%	18.894%	22.640%	16.185%

Latency Benchmarks (Single-Threaded)



Lower latency means faster operations. Measures the time (in nanoseconds) for individual Get, Set, and SetEvict operations in a single-threaded environment.

String Keys

Results

Cache	Get (ns)	Get allocs	Set (ns)	Set allocs	SetEvict (ns)	SetEvict allocs	Avg (ns)
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Cache	Get (ns)	Get allocs	Set (ns)	Set allocs	SetEvict (ns)	SetEvict allocs	Avg (ns)
fido	11.0	0	17.0	0	135.0	1	14.000
clock	18.0	0	25.0	0	111.0	2	21.500
s4lru	19.0	0	65.0	1	67.0	1	42.000
sieve	25.0	0	49.0	0	210.0	3	37.000
freelru- sync	26.0	0	25.0	0	39.0	0	25.500
freelru- shard	27.0	0	41.0	0	50.0	0	34.000
2q	30.0	0	32.0	0	257.0	2	31.000
lru	31.0	0	32.0	0	105.0	1	31.500
s3-fifo	33.0	0	49.0	0	401.0	5	41.000
otter	38.0	0	185.0	1	201.0	1	111.500
ristretto	65.0	1	285.0	3	152.0	4	175.000
tinylfu	91.0	0	129.0	3	134.0	3	110.000
freecache	95.0	1	67.0	0	176.0	0	81.000
theine	95.0	1	377.0	0	533.0	1	236.000
hypercache	116.0	1	237.0	2	1102.0	6	176.500
ttlcache	531.0	0	507.0	0	423.0	3	519.000

Int Keys

Results

Cache ▾	Get (ns) ▾	Get allocs ▾	Set (ns) ▾	Set allocs ▾	SetEvict (ns) ▾	SetEvict allocs ▾	Avg (ns) ▾
fido	8.0	0	15.0	0	113.0	1	11.500
freelru-shard	8.0	0	19.0	0	19.0	0	13.500
otter	33.0	0	187.0	1	189.0	1	110.000
theine	89.0	1	327.0	0	453.0	1	208.000
ttlcache	481.0	0	468.0	0	397.0	3	474.500

String Keys - GetOrSet

GetOrSet is an atomic operation that gets a value if it exists, or sets it if it doesn't. Only caches that support this operation are shown.

GetOrSet Results

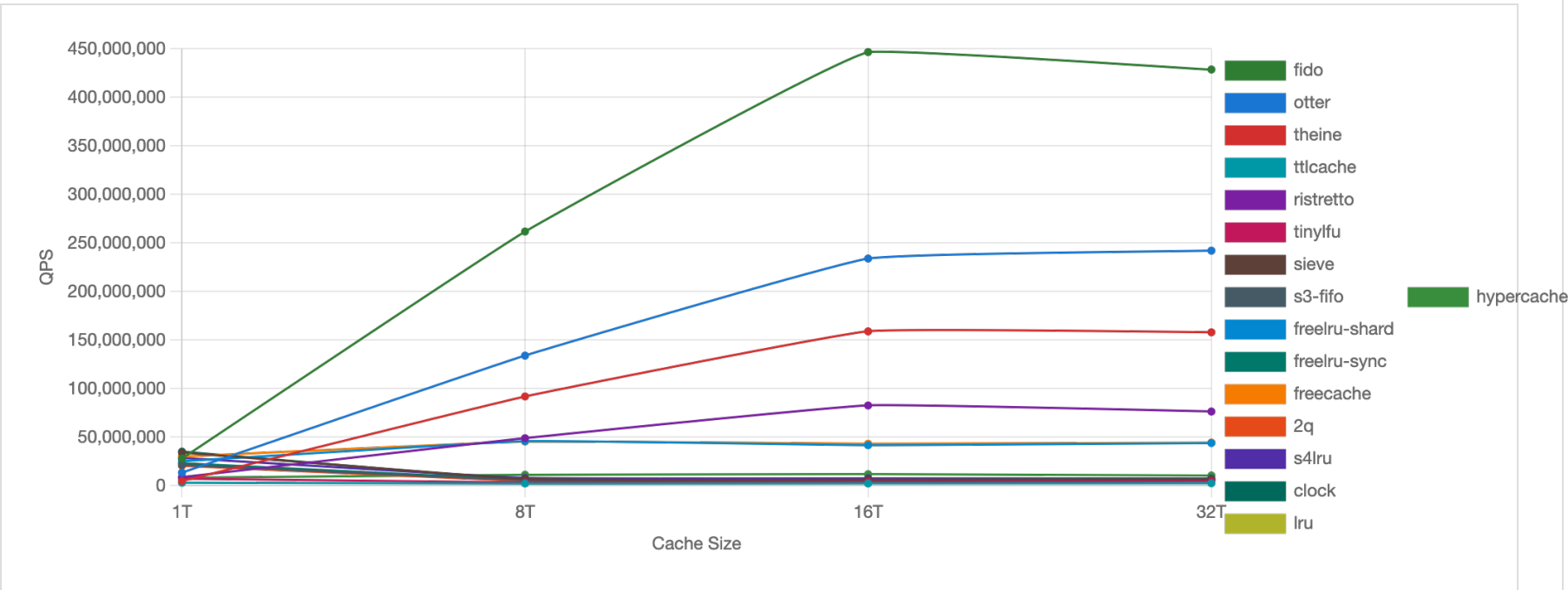
Cache ▾	GetOrSet (ns) ▾	GetOrSet allocs ▾
fido	13.0	0
otter	58.0	1
freecache	150.0	2
theine	190.0	4
hypercache	222.0	2
ttlcache	563.0	0

Throughput Benchmarks (Multi-Threaded)



Higher throughput means more queries per second (QPS). Tests measure concurrent Get and Set performance separately with a Zipf workload across different thread counts.

String Keys - Get

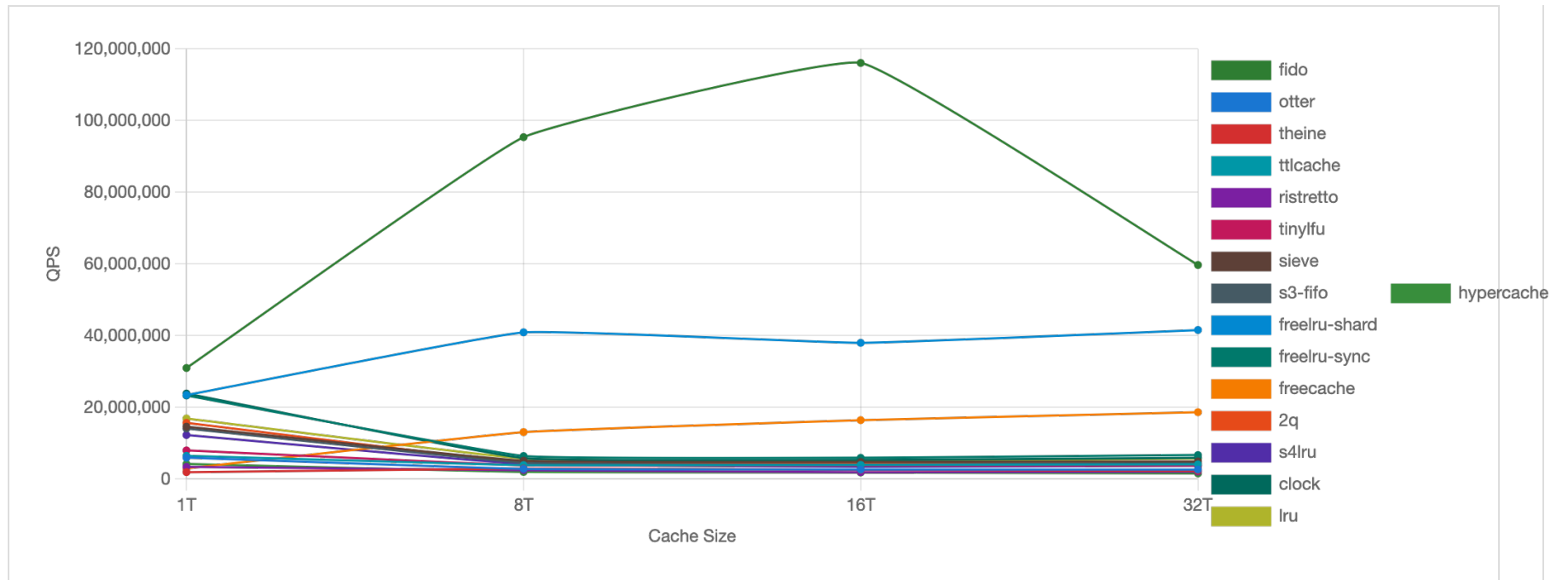


Results Table

Cache	1T	8T	16T	32T	Avg
fido	28.18M	261.88M	446.69M	428.66M	291.35M

Cache	1T	8T	16T	32T	Avg
otter	13.49M	134.10M	233.99M	242.11M	155.92M
theine	4.78M	91.89M	159.06M	158.05M	103.45M
ristretto	8.81M	49.01M	82.77M	76.51M	54.28M
freecache	30.17M	45.52M	43.33M	44.35M	40.84M
freelru-shard	25.33M	45.91M	41.87M	44.00M	39.28M
sieve	34.19M	6.96M	6.24M	6.60M	13.50M
clock	35.16M	6.18M	5.48M	5.76M	13.14M
s4lru	28.76M	8.16M	7.69M	7.56M	13.04M
freelru-sync	23.31M	6.71M	6.04M	7.20M	10.81M
hypercache	8.22M	11.21M	11.94M	10.46M	10.46M
lru	22.87M	5.39M	4.84M	5.94M	9.76M
s3-fifo	21.50M	5.95M	5.42M	5.91M	9.70M
2q	20.55M	5.11M	4.12M	5.56M	8.84M
tinylfu	7.26M	3.44M	3.96M	4.56M	4.81M
ttlcache	3.11M	2.31M	2.44M	2.59M	2.61M

String Keys - Set

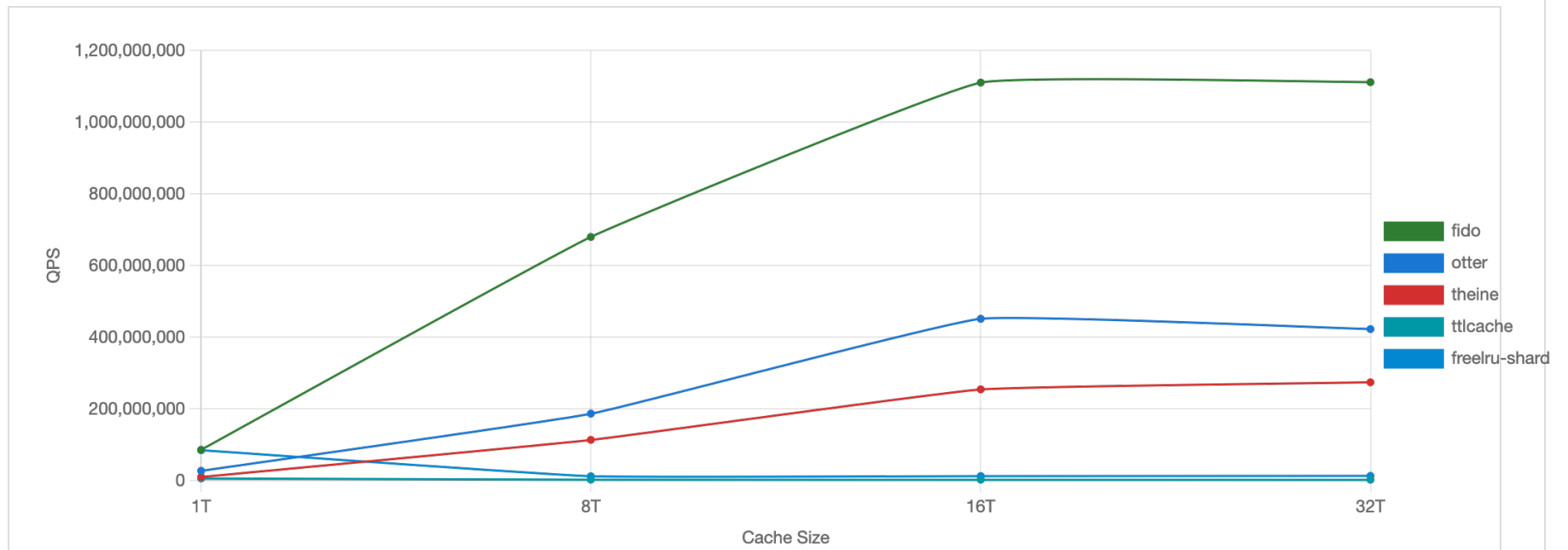


Results Table

Cache	1T	8T	16T	32T	Avg
fido	31.00M	95.40M	116.10M	59.73M	75.56M
freelru-shard	23.46M	40.94M	38.03M	41.58M	36.00M
freecache	2.98M	13.09M	16.43M	18.66M	12.79M
freelru-sync	23.30M	6.44M	5.95M	6.71M	10.60M
clock	23.88M	5.86M	5.39M	5.93M	10.26M
lru	16.89M	5.73M	4.84M	5.75M	8.30M
sieve	14.62M	5.12M	4.80M	4.94M	7.37M
2q	15.67M	4.63M	4.38M	4.23M	7.23M

Cache	1T	8T	16T	32T	Avg
s3-fifo	14.13M	4.58M	4.01M	4.55M	6.82M
s4lru	12.30M	4.23M	3.45M	3.72M	5.92M
tinylfu	8.03M	3.88M	3.83M	3.84M	4.89M
ttlcache	6.50M	3.91M	3.96M	4.13M	4.62M
otter	5.99M	2.84M	2.67M	2.65M	3.54M
ristretto	3.43M	2.49M	1.94M	2.08M	2.48M
theine	1.94M	2.94M	2.58M	2.33M	2.45M
hypercache	4.22M	2.02M	1.84M	1.60M	2.42M

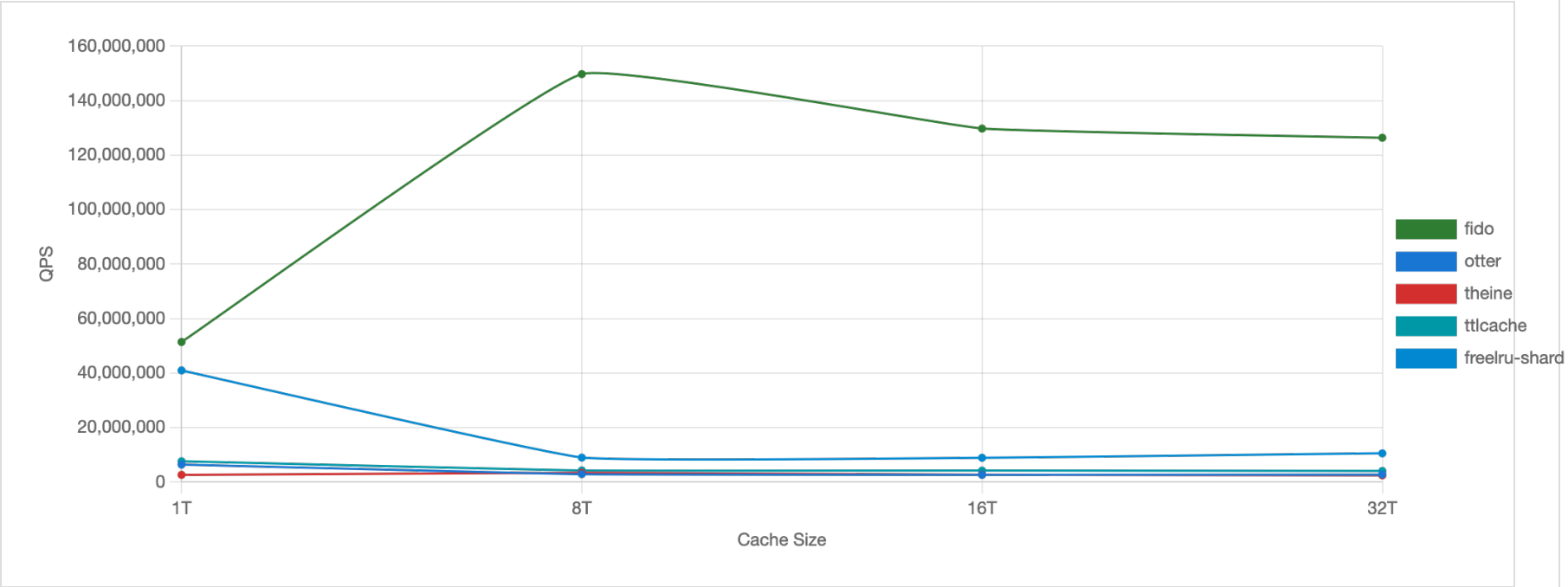
Int Keys - Get



Results Table

Cache	1T	8T	16T	32T	Avg
fido	86.18M	680.17M	1110.68M	1111.86M	747.22M
otter	27.53M	186.92M	451.91M	423.16M	272.38M
theine	10.57M	113.90M	254.85M	274.79M	163.53M
freelru-shard	85.40M	12.63M	12.95M	13.51M	31.12M
ttlcache	6.58M	3.07M	2.77M	2.70M	3.78M

Int Keys - Set

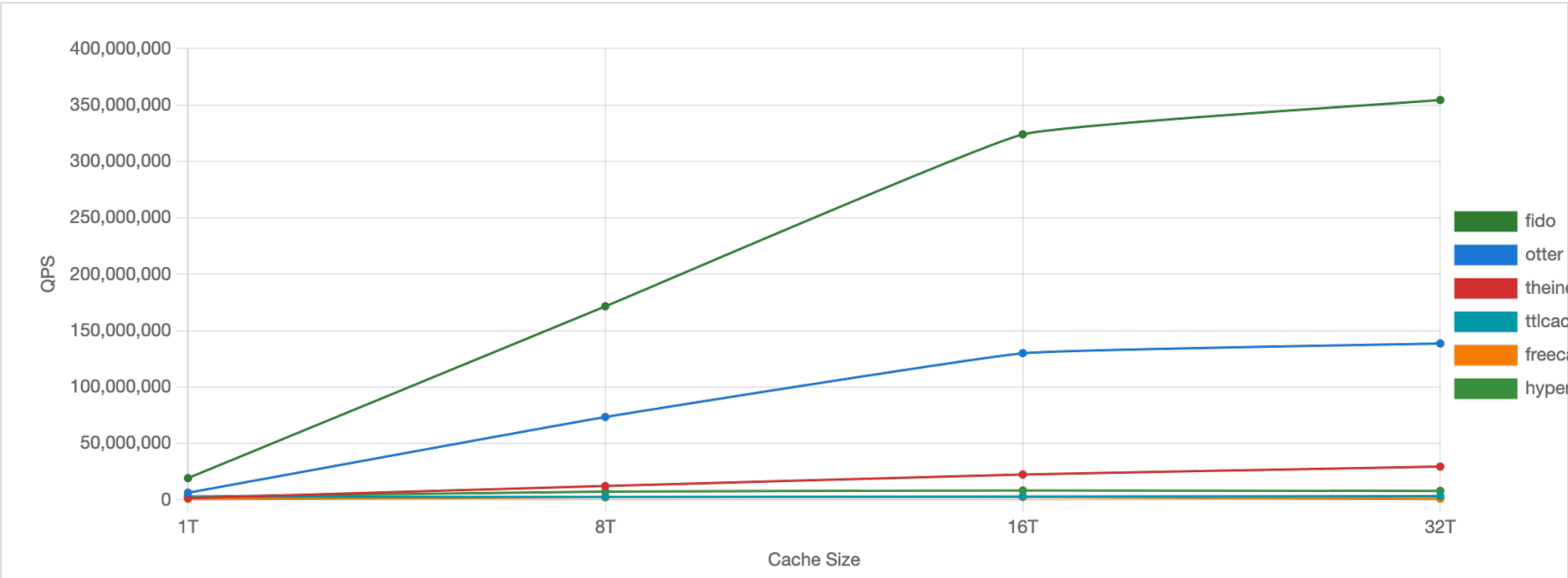


Results Table

Cache	1T	8T	16T	32T	Avg
fido	51.42M	149.71M	129.77M	126.35M	114.31M
freelru-shard	40.95M	9.02M	8.90M	10.57M	17.36M
ttlcache	7.63M	4.25M	4.23M	4.10M	5.05M
otter	6.46M	2.93M	2.69M	2.77M	3.71M
theine	2.67M	3.42M	2.73M	2.49M	2.83M

GetOrSet Operations

Atomic get-or-compute operations with URL-like keys. Only caches supporting GetOrSet are shown.



Results Table

Cache	1T	8T	16T	32T	Avg
fido	19.31M	171.57M	323.95M	354.44M	217.31M
otter	6.39M	73.55M	130.03M	138.72M	87.17M
theine	1.60M	12.30M	22.47M	29.53M	16.48M
hypercache	2.83M	7.33M	8.35M	7.93M	6.61M
ttlcache	2.82M	2.64M	2.92M	3.25M	2.91M
freecache	950K	2.48M	2.55M	1.07M	1.76M

Memory Overhead Benchmarks



Capacity: 32768 items, Value size: 1024 bytes. Measured in isolated processes. Overhead compared to baseline map[string][]byte. Lower is better.

Memory Results

Cache	Items Stored	Memory (MB)	Overhead vs map (bytes/item)
freelru-sync	32768	35.32	-20
otter	32768	36.45	15

Cache	Items Stored	Memory (MB)	Overhead vs map (bytes/item)
freelru-shard	31606	36.60	■ 21
clock	32768	37.13	■ 37
fido	32768	37.51	■ 49
2q	32768	37.63	■ 53
lru	32768	37.66	■ 54
s4lru	32768	38.39	■ 77
theine	32768	38.60	■ 84
sieve	32768	39.66	■ 118
s3-fifo	32768	39.66	■ 118
tinylfu	32767	39.96	■ 127
ttlcache	32768	40.94	■ 159
ristretto	32768	41.60	■ 180
hypercache	32768	43.39	■ 237
freecache	32764	44.90	■ 285